

Neuro-Glove Rehabilitation Suite

Comprehensive 9-Game Clinical Protocol and Exercise Mapping Guide

PROTOCOL SUITE	HARDWARE PROFILE	INPUT MODALITY	CLINICAL TARGET
9 Therapeutic Games	StretchSense Wireless Glove	Pure Sensor Telemetry (100%)	Upper Limb Neuro-Motor Recovery

1. Executive Clinical Protocol Matrix

The Neuro-Glove Rehabilitation Suite translates therapeutic upper-extremity motor rehabilitation into interactive, biofeedback-driven 3D environments. Each game directly maps to an isolated anatomical motion protocol or compound agility workflow. All patient actions are executed strictly through natural hand, finger, and wrist articulations without keyboard or mouse requirements.

Game Title	Rehabilitation Exercise	Primary Biomechanical Motions	Target Pathologies	Sensor Telemetry
Gelato Tower	Forearm Pronation / Supination	Forearm axial rotation around longitudinal axis; cone tilt balance	Stroke spasticity, Colles fracture stiffness, pronator synergy	Roll IMU (-45° to +45°)
Sky Glider	Forearm Pronation / Supination	Aileron flight banking; forearm rotation steers aircraft through rings	Rotational arc deficit, radial head trauma, hemiparesis	Roll IMU (-45° to +45°)
Highway Racer	Forearm Pronation / Supination	Automotive steering column; palm down/up triggers lane shifts	Dynamic rotational control, driving re-education, synergy suppression	Roll zones (<-15°, 0°, >+15°)
Balloon Blitz	Hand Open / Close	Pneumatic pump grasp; full digital extension balloon launch	Radial nerve palsy, extensor lag, grasp-release deficit	5-ch bend sensors (0-100%)
Claw Crane	Hand Open / Close	Claw clamp grasp; open hand active release over transport hold	Flexor hypertonia, release timing deficit, grasp calibration	Flexion (>55% / <25%)
Hill Climb 4x4	Hand Open / Close	Power grip engine throttle; open hand active brake caliper control	Sustained isometric grip fatigue, graded release modulation	Continuous flexion (0-100%)
Pizza Spin	Wrist Circumduction	Continuous 360° multi-planar carpal circle spinning pizza dough	Carpal tunnel post-op, radiocarpal contracture, limited arc	Carpal vector (0° to 360°)
Captain's Wheel	Wrist Circumduction	Continuous 360° wrist circle revolving nautical brass ship helm	Chronic carpal stiffness, multi-axial articular excursion	Circular accumulator (0°-360°)
Rolling Wonder	Dynamic Compound Agility	Wrist tilt steering, finger extension leaping, 360° circumduction turbo	Complex motor coordination, cognitive-motor dual tasking	Integrated multi-sensor bridge

2. Architecture & Sensor Telemetry Pipeline

Direct Bluetooth Telemetry Bridge: The StretchSense rehabilitation glove streams continuous high-frequency digital packets directly into the platform bridge via Web Bluetooth / WebSocket telemetry. Data streams include 5 discrete digital bend sensors (thumb, index, middle, ring, little) and a 6-DoF inertial measurement unit (IMU).

Zero-Latency Procedural Feedback: Telemetry updates render at 60 frames per second directly in WebGL canvas contexts, accompanied by reactive WebAudio synthesizers generating auditory confirmation for range thresholds, grasp limits, and level milestones.

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PROTOCOL 1: FOREARM PRONATION & SUPINATION

MODULE: GELATO TOWER

Gelato Tower: Rotational Balance & Dynamic Stacking

Clinical View: Waffle cone tilt tracking via continuous forearm pronation and supination telemetry.

CLINICAL RATIONALE

Forearm rotation is crucial for activities of daily living (turning keys, holding bowls, using utensils). Gelato Tower encourages active rotational range of motion (ROM) through fine motor counter-balancing, breaking pathological pronator synergy patterns in stroke survivors.

TARGETED MUSCLE GROUPS

Pronation: Pronator teres, pronator quadratus.

Supination: Supinator, biceps brachii, brachioradialis.

HOW TO PLAY WITH GLOVES

- **Posture:** Elbow flexed to 90° resting on table; vertical thumb-up establishes neutral (0°).
- **Pronation (Tilt Left):** Rotate palm downward to angle cone left and catch left scoops.
- **Supination (Tilt Right):** Rotate palm upward to angle cone right and catch right scoops.
- **Neutral Return:** Return to vertical thumb-up to stabilize stack center of gravity.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

TIERS 1-2 Vanilla Coast & Pistachio Park: Slow fall speed, generous catch basin; targets 8 and 12 scoops.

TIERS 3-4 Berry Mountain & Golden Sunset: Moderate stack sway requiring fine-motor counter-rotations; targets 16 and 20 scoops.

TIER 5 Grand Gelateria: Rapid alternate-quadrant spawns demanding rapid rotational alternation; target: 25 scoops.

Clinical Objective: Expand active rotational arc from $\pm 15^\circ$ in early stages toward full functional $\pm 45^\circ$ bilateral excursion.

Sky Glider: Aileron Flight Banking & Canyon Navigation



Clinical View: Glider roll ailerons actuated directly by forearm rotation through golden canyon rings.

CLINICAL RATIONALE

Sky Glider utilizes an intuitive aerodynamic flight metaphor: patient forearm rotation directly mirrors aircraft roll ailerons. This reinforces rotational motor intention, encouraging sustained, smooth pronation/supination excursions to traverse canyon target rings.

TARGETED MUSCLE GROUPS

Pronation (Left Wing Dip): Pronator quadratus, pronator teres.
Supination (Right Wing Dip): Supinator, biceps brachii.

HOW TO PLAY WITH GLOVES

- **Posture:** Forearm extended forward resting on a support wedge in neutral mid-position.
- **Pronation Banking (Fly Left):** Rotate palm downward. The glider rolls left, banking into the left corridor.
- **Supination Banking (Fly Right):** Rotate palm upward. The glider rolls right, gliding into the right corridor.
- **Neutral Glide:** Return to vertical alignment to level wings and hold steady course.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

- TIERS 1-2 Breeze Valley & Emerald Gorge:** Broad canyon paths, gentle ring placement; targets 8 and 12 rings.
- TIERS 3-4 Azure Peaks & Sunset Ridge:** Tighter ring gates requiring continuous active rotational steering; targets 16 and 20 rings.
- TIER 5 Grand Canyon:** Narrow rock arches and rapid alternating slalom gates; target: 25 rings.

Clinical Objective: Train smooth, non-jerky rotational velocity and precise angular mid-range holding.

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PROTOCOL 1: FOREARM PRONATION & SUPINATION

MODULE: HIGHWAY RACER

Highway Racer: Steering Column & Triple Lane Shifts

Clinical View: Vehicle steering column rotation executing left, center, and right freeway lane changes.

CLINICAL RATIONALE

Automobile steering demands discrete, decisive forearm rotational transitions with an immediate return to neutral stability. Highway Racer casts the forearm as an automotive steering column, training rapid motor switching between discrete rotational thresholds.

TARGETED MUSCLE GROUPS

Pronation (Left Lane Change): Pronator teres, pronator quadratus.

Supination (Right Lane Change): Supinator, biceps brachii.

Co-Contraction: Stabilizers maintaining center deadband.

HOW TO PLAY WITH GLOVES

- **Posture:** Forearm elevated in driving posture, mimicking grasping a steering wheel.
- **Pronation (Shift Left):** Rotate palm downward past -15° threshold to shift car into left highway lane.
- **Supination (Shift Right):** Rotate palm upward past $+15^\circ$ threshold to shift car into right highway lane.
- **Neutral Deadband:** Hold forearm within -15° to $+15^\circ$ zone to cruise steadily in center lane.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

TIERS 1-2 Coastal Highway & Desert Freeway: Wide 3-lane road, cruising speed 60-75 km/h; targets 200m and 300m.

TIERS 3-4 Alpine Expressway & Sunset Boulevard: Speed increases to 90 km/h, closer traffic pacing; targets 400m and 500m.

TIER 5 Grand Prix Circuit: Peak speed 120 km/h, rapid lane shifts between obstacles; target: 650m.

Clinical Objective: Train cognitive-motor reaction time and rapid agonist-antagonist forearm reciprocal inhibition.

Balloon Blitz: Pneumatic Grasp & Open Hand Launch



Clinical View: Graded fist squeeze pumping air; maximal digital extension launching target balloons.

CLINICAL RATIONALE

Rehabilitating functional hand grasp demands alternating between full digital flexion (power grasp) and active digital extension (release). Balloon Blitz couples fist closure to pneumatic inflation, while rewarding complete finger release with ascending balloon launches.

TARGETED MUSCLE GROUPS

Digital Flexion (Grasp Pump): Flexor digitorum superficialis, flexor digitorum profundus, flexor pollicis longus.

Digital Extension (Launch): Extensor digitorum communis, extensor indicis, extensor digiti minimi.

HOW TO PLAY WITH GLOVES

- **Posture:** Forearm flat on table, fingers resting loosely in comfortable semi-flexion.
- **Close Fist (Grasp to Pump):** Squeeze all fingers into a firm fist (>55% bend) to actuate the red pneumatic piston.
- **Target Green Window:** Pump until volume enters optimal 70%-85% zone. Avoid bursting (>95%).
- **Open Hand (Release to Launch):** Spread all five fingers wide into full extension (<25% bend) to launch.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

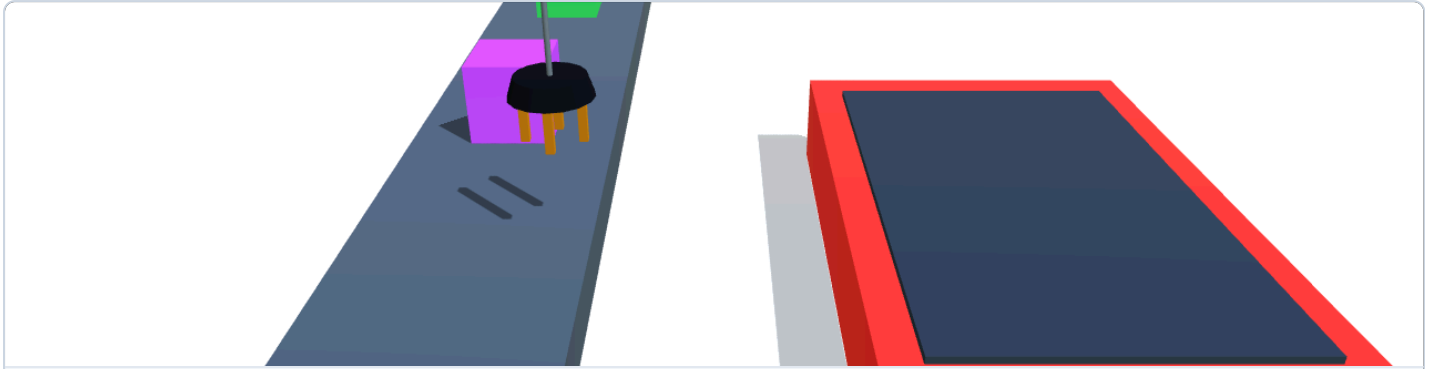
TIERS 1-2 Meadow Carnival & Spring Fair: Wide target green zone (65%-85%), stable pressure; targets 6 and 9 balloons.

TIERS 3-4 Skyway Fiesta & Twilight Bazaar: Narrower target window (72%-84%), sensitive pump pacing; targets 12 and 15 balloons.

TIER 5 Grand Gala: Precision window (75%-83%), strict over-inflation burst limit; target: 18 balloons.

Clinical Objective: Overcome extensor lag and cultivate voluntary inhibitory control against spastic over-clenching.

Claw Crane: Robotic Clamp Squeeze & Active Cargo Drop



Clinical View: 4-prong crane claw directly actuated by patient fingers, picking cargo and dropping into ship hold.

CLINICAL RATIONALE

Claw Crane provides an explicit biomechanical mirror: the patient's fingers physically articulate the 4 prongs of an overhead cargo crane. Squeezing clamps the claw prongs around crates; opening the hand releases cargo into a ship hold, validating grip endurance and release timing.

TARGETED MUSCLE GROUPS

Digital Flexion (Claw Clamping): Flexor digitorum superficialis, profundus, thenar intrinsics.

Digital Extension (Claw Opening): Extensor digitorum communis, abductor pollicis longus.

HOW TO PLAY WITH GLOVES

- **Posture:** Forearm supported with wrist in slight functional extension (15°-20°).
- **Close Fist (Clamp Claw):** Clench fingers (>55% bend) when crate is in pickup zone to clamp claw prongs.
- **Hold Grasp:** Maintain closed fingers while crane trolley traverses across conveyor to ship.
- **Open Hand (Drop Cargo):** Open all five fingers wide (<25% bend) over target hold beacon to release crate.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

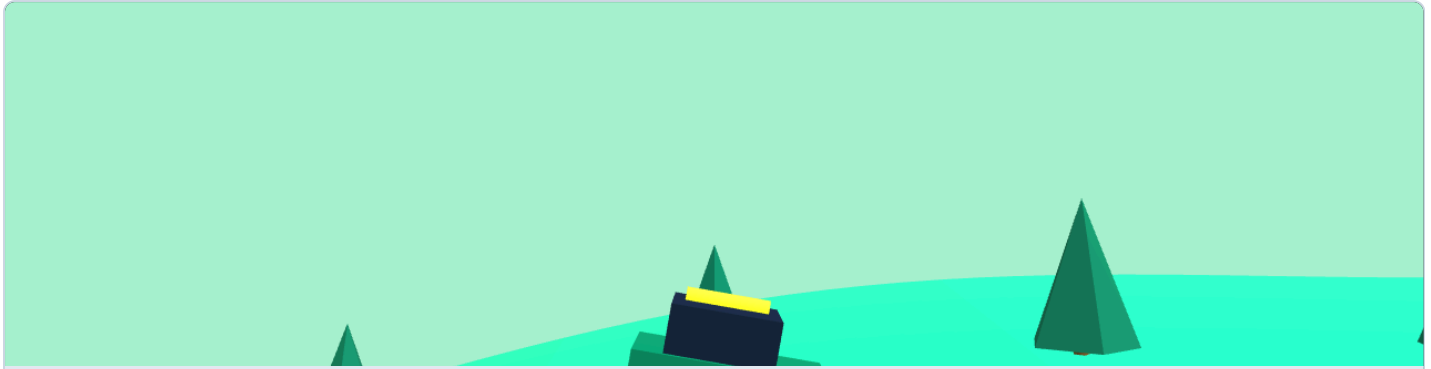
TIERS 1-2 Dockside Intake & Harbor Transfer: Slow conveyor, large ship target zone; targets 6 and 9 crates.

TIERS 3-4 Container Port & Deep Sea Terminal: Faster conveyor pacing, tighter drop tolerances; targets 12 and 15 crates.

TIER 5 Global Freight Hub: Rapid continuous conveyor feed requiring rapid grasp-release cycles; target: 18 crates.

Clinical Objective: Strengthen sustained isometric finger flexion during transit and eliminate extensor release delay.

Hill Climb 4x4: Grip Power Throttle & Active Release Braking



Clinical View: Fist squeeze delivers engine torque; open hand activates brake calipers across rolling terrain.

CLINICAL RATIONALE

Hill Climb 4x4 trains graded hand grasp modulation and rapid extensor braking. Climbing rolling alpine peaks requires patients to modulate isometric grip strength to control torque, releasing into active extension to brake before downhill crests.

TARGETED MUSCLE GROUPS

Hand Grasp (Torque Throttle): Flexor digitorum profundus, flexor digitorum superficialis, interossei.

Hand Extension (Braking): Extensor digitorum, extensor pollicis longus.

HOW TO PLAY WITH GLOVES

- **Posture:** Forearm comfortably supported on therapy table, hand ready in neutral posture.
- **Squeeze Fist (Throttle):** Clench fist (>50% bend). Throttle bar fills green, accelerating the jeep up hills.
- **Open Hand (Braking):** Spread fingers into wide extension (<30% bend) to apply brakes before crest drop-offs.
- **Relaxed Hand (Coasting):** Rest fingers in mid-range (30%-50%) for momentum coasting.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

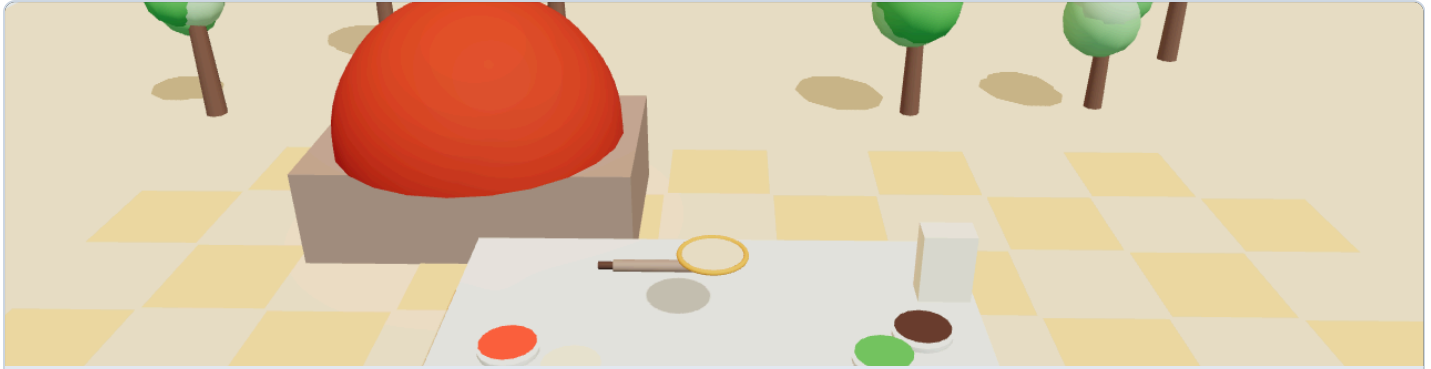
TIERS 1-2 Green Foothills & Rocky Bluffs: Rolling gentle peaks, base speeds 45-50 km/h; targets 200m and 300m.

TIERS 3-4 Pine Ridge & Canyon Crossing: Frequent steep inclines requiring rapid throttle-brake alternation; targets 400m and 500m.

TIER 5 Everest Summit: Extreme alpine slopes demanding peak isometric grip endurance; target: 650m.

Clinical Objective: Train graded muscular force gradation rather than all-or-nothing spastic clutching.

Pizza Spin: Carpal Articular Excursion & Dough Expansion



Clinical View: Full 360° circular wrist circumduction expanding artisanal pizza dough on a wooden peel.

CLINICAL RATIONALE

Wrist circumduction is a multi-planar composite motion combining flexion, extension, radial deviation, and ulnar deviation. Following wrist immobilization or carpal tunnel decompression, patients frequently develop restricted, angular motions. Pizza Spin trains fluid circular articulation.

TARGETED MUSCLE GROUPS

Carpal Flexors/Extensors: Flexor carpi radialis, flexor carpi ulnaris, extensor carpi radialis longus/brevis, extensor carpi ulnaris.

Articular Circle: Seamless sequential kinetic hand-off.

HOW TO PLAY WITH GLOVES

- **Posture:** Forearm flat on table, wrist joint positioned freely off edge or on a soft elevation pad.
- **Execute Smooth Circles:** Guide hand in a continuous 360° circular perimeter at the wrist without elbow compensation.
- **Sequential Sequence:** Upward extension → radial deviation → downward flexion → ulnar deviation.
- **Dough Expansion:** Each completed 360° circle expands pizza dough diameter toward 14-inch goal.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

TIERS 1-2 Trattoria Rustica & Napoli Classic: Targets 12 and 14 inch crusts (requires 4 and 6 completed circles).

TIERS 3-4 Firenze Gourmet & Venezia Master: Targets 16 and 18 inch crusts (requires 8 and 10 completed circles).

TIER 5 Grand Pizzeria: Target 20 inch crust (requires 12 continuous completed circles).

Clinical Objective: Restore smooth multi-axial articular excursion and reduce carpal joint stiffness.

Captain's Wheel: Nautical Helm & 360° Ocean Navigation



Clinical View: Continuous 360° wrist circular revolution turning brass ship wheel to navigate ocean reefs.

CLINICAL RATIONALE

Captain's Wheel provides an organic nautical metaphor: patient circular wrist motion directly turns an authentic brass-fitted wooden ship helm, steering an ocean galleon around coral reefs and collecting sunken treasure chests.

TARGETED MUSCLE GROUPS

Circumduction Musculature: Coordinated co-activation of flexor carpi radialis, flexor carpi ulnaris, extensor carpi radialis longus, extensor carpi ulnaris.

Multi-Planar Endurance: Sustained rotational pacing.

HOW TO PLAY WITH GLOVES

- **Posture:** Forearm resting on therapy armrest, hand and wrist suspended freely in mid-air.
- **Rotate Wrist 360°:** Revolve hand in wide, continuous full circles at the carpal joint.
- **Turn Ship Helm:** The 3D ship wheel spins synchronously with the patient's wrist trajectory.
- **Steer & Salvage:** Directional circles steer ship through open channels, salvaging floating gold chests.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

TIERS 1-2 **Calm Shoals & Coral Lagoon:** Gentle open water, slow reef drifts; targets 4 and 6 helm rotations.

TIERS 3-4 **Pirate Cove & Stormy Strait:** Narrow island channels, ocean swells; targets 8 and 10 helm rotations.

TIER 5 **Open Ocean Odyssey:** High-seas expedition requiring persistent circular pacing; target: 12 helm rotations.

Clinical Objective: Rebuild sustained multi-axial carpal endurance and prevent joint contracture.

Rolling Wonder: 3D Wrist Circumduction Runner (Pip's Meadow)



Clinical View: Multi-modal runner integrating lateral tilt steering, extension jumping, and circumduction turbo.

CLINICAL RATIONALE

Functional motor autonomy requires synthesizing isolated joint motions into compound coordination. Rolling Wonder combines lateral wrist tilt steering, rapid hand extension hurdle leaps, and 360° circumduction turbo propulsion into a continuous runner.

TARGETED MUSCLE GROUPS

Compound Coordination: Forearm pronators/supinators (steering), digital extensors (jumping), and carpal circumductors (rainbow turbo boost).

HOW TO PLAY WITH GLOVES

- **Clockwise Rotation (Turn Right):** Smooth clockwise wrist circumduction steers Pip toward right track lanes (Registers Step 1: Turn Right ✓).
- **Anticlockwise Rotation (Turn Left):** Reverse wrist into anticlockwise circumduction to steer Pip left, completing **1 Full Clinical Repetition** (Turn Right → Turn Left).
- **Hand Open Leap:** Snap all five fingers wide into full extension (<25% bend) to leap over obstacles and catch star rings.

BIOFEEDBACK & PROGRESSIVE DIFFICULTY TIERS

- TIERS 1-2 Sunny Meadow & Candy Canyon:** 120m-150m courses; targets 6 and 8 completed circumduction reps (Right → Left).
- TIERS 3-4 Rainbow Skyway & Carnival Coaster:** 180m-210m courses; targets 10 and 12 completed circumduction reps.
- TIER 5 Grand Champion Tour:** 250m championship course; target: 15 completed circumduction reps.

Clinical Objective: Promote cognitive-motor dual tasking, motor anticipation, and dynamic limb stabilization.

1. Structured Weekly Clinical Curriculum

To maximize neuroplastic reorganization while mitigating muscle fatigue, clinicians should structure weekly sessions according to targeted movement domains:

Day	Target Domain	Prescribed Game Modules	Recommended Clinical Dosage
Monday	Forearm Pronation / Supination	Gelato Tower & Sky Glider	3 sets of 12 scoops caught / 12 flight rings
Tuesday	Grasp & Active Digital Extension	Balloon Blitz & Claw Crane	3 sets of 10 launched balloons / 10 crane drops
Wednesday	Carpal Circumduction Mobility	Pizza Spin & Captain's Wheel	4 sets of 8 completed 360° circles
Thursday	Power Modulation & Sustained Grip	Hill Climb 4×4 & Highway Racer	3 runs of 300m alpine ascent / freeway course
Friday	Compound Agility & Evaluation	Rolling Wonder & Full Suite Check	2 runs of 200m compound runner + clinical audit

2. Clinical Administration Guidelines

PRE-SESSION TARE & BASELINE CALIBRATION

Ensure the patient assumes a comfortable resting neutral posture before session launch. Confirm through the connection modal that baseline bend sensors read below 20% and IMU roll angle is centered at 0° (±2°).

GUARDING AGAINST SYNERGISTIC COMPENSATION

Patients with upper-extremity hemiparesis frequently compensate for restricted forearm supination by abducting the shoulder or flexing the torso laterally. Keep the elbow tucked near 90° flexion, utilizing an arm trough support if unprompted shoulder hiking appears.

MONITORING SPASTIC FATIGUE & REST INTERVALS

Enforce 30-to-60 second rest pauses between level completions. If flexor spasticity increases or active extensor release latency exceeds 2.5 seconds, pause the module and perform gentle passive extensor stretching.